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DISPENSER AND LIFTING DEVICE OF DISPENSER

Abstract

The present disclosure provides a dispenser for dispensing a material on a substrate. The dispenser includes a pair of rails, a heating device, and a pair of lifting devices. One of the pair of rails is able to move towards or away from the other in the Y-axis direction to adjust the distance between the pair of rails, each rail having a supporting surface and a blocking portion above the supporting surface for supporting the substrate in a transport position, the blocking portion being used to keep the substrate in a dispensing position. The heating device is located below the pair of rails in the Z-axis direction and is movable in the Z-axis direction. The pair of lifting devices are respectively mounted on the pair of rails, each including: a lifting member movably connected to the rail and a reset member between the rail and the lifting member. The lifting member is able to be raised under pushing of the heating device to cause the substrate to rise from the transport position to the dispensing position and to lower the substrate from the dispensing position to the transport position under the action of the reset member when the heating device descends.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure generally relates to a device for dispensing a viscous material on a substrate (such as a printed circuit board).

BACKGROUND ART

[0002] During manufacturing of printed circuit boards, in some cases, there is a need to dispense a viscous material (e.g., an viscous) on the circuit boards for better connecting some circuit elements to the circuit boards. A dispenser for dispensing the viscous material on the printed circuit board is also referred to as a dispenser. In such a dispenser, there is a transport rail for transporting the substrate, a dispensing unit located above the rail, and a heating device located below the rail. The heating device is used to heat the substrate during dispensing operations by the dispensing unit so that the viscous material is better attached to the desired position of the substrate.

SUMMARY OF THE INVENTION

[0003] According to a first aspect of the present disclosure, there is provided a dispenser for dispensing a material on a substrate. The dispenser includes: a pair of rails configured to input and output the substrate to the dispenser in an X-axis direction, where one of the pair of rails is movable in a Y-axis direction towards or away from the other to adjust a distance between the pair of rails, a supporting surface and a blocking portion higher than the supporting surface are disposed on each of the pairs of rails, the supporting surface of the pair of rails is configured to support the substrate in a transport position, and the blocking portion of the pair of rails is configured to keep the substrate in a dispensing position; a heating device located below the pair of rails in a Z-axis direction and movable in the Z-axis direction; and a pair of lifting devices mounted to the pair of rails respectively, each of the pair of lifting devices including: a lifting member movably connected to the rail and movable in the Z-axis direction; and a reset member located between the rail and the lifting member to be compressed when the lifting member is raised relative to the rail and to be able to lower the lifting member relative to the rail by a resilient force; where the lifting member is configured to be able to be raised under pushing of the heating device to cause the substrate to rise from the transport position to the dispensing position, and to lower the substrate to the transport position from the dispensing position under an action of the reset member when the heating device descends.

[0004] According to the dispenser described above, the lifting device further includes a connecting device including a connecting slot disposed on the lifting member and a connecting member connected to the rail, the connecting slot extending along the Z-axis direction, the connecting member being inserted into the connecting slot and being movable in the connecting slot in the Z-axis direction.

[0005] According to the dispenser described above, the lifting member is a clevis-like and sleeved on the rail from the underside of the rail, the lifting member including a pair of side plates and a base plate connected between the pair of side plates, the pair of side plates being respectively located on opposite sides of the rail, the base plate being located below the rail, where the pair of side plates each has at least one connecting slot.

[0006] According to the dispenser described above, the lifting member extends by a distance along the X-axis direction. The lifting device includes several reset members, several connecting slots

disposed on each of the side plates, and several connecting members corresponding to the several connecting slots, where the several reset members are arranged in the X-axis direction, and the several connecting slots are arranged in the X-axis direction.

[0007] According to the dispenser described above, a hole is provided in the rail, and the reset member is at least partially located in the hole.

[0008] According to the dispenser described above, the reset member is a helical spring.

[0009] According to the dispenser described above, the dispenser also includes a guide pin, a bottom end of the guide pin is connected to the lifting member, and the reset member is sleeved on the guide pin.

[0010] According to the dispenser described above, the lifting device further includes a guide wheel having a slit on an exterior surface thereof, the slit extending circumferentially along the guide wheel. The guide wheel is sleeved on the connecting member, the guide wheel is inserted into the connecting slot, and opposite edges of the side plate of the lifting member forming the connecting slot are clamped in the slit of the guide wheel.

[0011] According to the dispenser described above, the blocking portion is formed by a baffle mounted at a top end of the rail.

[0012] According to the dispenser described above, the dispenser further includes a drive device disposed below the heating device for driving movement of the heating device in the Z-axis direction.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. 1 is a stereoscopic view of a dispenser according to an example of the present disclosure;

[0014] FIG. 2 is a stereoscopic view of a rail, a lifting device, a heating device, and a drive device of FIG. 1;

[0015] FIG. 3A is a stereoscopic view of the rail and the lifting device of FIG. 2;

[0016] FIG. 3B is an exploded view of the rail and the lifting device of FIG. 2;

[0017] FIG. 3C is a partial enlarged view of a top view of the rail and the lifting device of FIG. 2;

[0018] FIG. 3D is a cross-sectional view along line A-A in FIG. 3C;

[0019] FIG. 3E is a cross-sectional view along line B-B in FIG. 3C;

[0020] FIG. 4A is a cross-sectional view of a substrate of FIG. 2 in a transport position;

[0021] FIG. 4B is a cross-sectional view of the heating device of FIG. 2 when raised to contact the lifting device; and

[0022] FIG. 4C is a cross-sectional view of the substrate of FIG. 2 when raised to a dispensing position.

DESCRIPTION OF EMBODIMENTS

[0023] Various specific embodiments of the present disclosure will be described below with reference to the appended drawings that form a part of the present specification. It should be understood that while terms denoting orientation, such as “front”, “rear”, “upper”, “lower”, “left”, “right”, “top”, “bottom”, etc., are used in the present disclosure to describe various exemplary structural parts and elements of the present disclosure, these terms are used herein for convenience of illustration only and are determined based on the exemplary orientations shown in the appended drawings. Since the examples disclosed in the present disclosure may be disposed in different orientations, these terms denoting orientation are for illustrative purposes only and should not be considered as limiting.

[0024] FIG. 1 is a stereoscopic view of a dispenser **100** according to an example of the present disclosure. As shown in FIG. 1, the dispenser **100** includes a body frame **102** that is box-like with a

width extending in the X-axis direction, a length extending in the Y-axis direction, and a height extending in the Z-axis direction. The body frame **102** has a housing space for supporting and housing various components of the dispenser **100**. The dispenser **100** also includes a front cover **103** disposed on the body frame **102** for opening and closing the housing space of the body frame **102**. The body frame **102** includes an opening **107** that run through itself in the X-axis direction. The dispenser **100** includes a pair of rails **120** that extend into the body frame **102** via the opening **107** for inputting and outputting a substrate **110** (e.g., a printed circuit board) into or out of the dispenser **100** for dispensing a viscous material (e.g., a binder) onto the substrate **110**. The dispenser **100** also includes a dispensing unit **105** disposed above the rail **120** and a rack **150** for the dispensing unit **105**. The dispensing unit **105** is, for example, a dispensing head, which is capable of dispensing a viscous material onto the substrate **110**. The quantity of dispensing units **105** may be set as desired, and two dispensing units are side by side in the X-axis direction in the example shown in FIG. 1. The rack **150** is used to move the dispensing unit **105** in the Y-axis direction and the X-axis direction to move the dispensing unit **105** over the position to be dispensed of the substrate **110**. The dispenser **120** also includes a heating device **201** disposed below the rail **120** for heating the substrate **110** prior to the dispensing unit **105** performing a dispensing operation, so as to better adhere the dispensed viscous material onto the substrate **110**.

[0025] In the dispenser **100** according to the present disclosure, the dispensing unit **105** is able to move in the Y-axis and X-axis directions parallel to the rail **120** to move to a desired dispensing position on the alignment substrate **110** prior to performing the dispensing operation, but the dispensing unit **105** does not move in the Z-axis direction perpendicular to the rail to move near the dispensing position on the substrate, but rather drives the substrate **110** up through the lifting heating device **201** to cause the substrate **110** to be proximate to the dispensing unit **105**. Therefore, the dispenser **100** of the present disclosure also includes a drive device **203** for driving movement of the heating device **201** in the Z-axis direction (as shown in FIG. 2) and a pair of lifting devices **250** (as shown in FIG. 2) mounted respectively on the pair of rails **120**.

[0026] FIG. 2 is a stereoscopic view of the rail **120**, the lifting device **250**, the heating device **201**, and the drive device **203** of FIG. 1. As shown in FIG. 2, the pair of rails **120** are disposed in parallel with each other and extend in the X-axis direction. One of the rails is a fixed rail and the other is a movable rail, which is movable in the Y-axis direction. Accordingly, the distance between the pair of rails **120** may be adjusted to accommodate different substrate widths. Each rail **120** is provided with a transport belt **208**, such as an annular belt, for transporting the substrate **110** to move along the X-axis direction. The transport belt **208** is disposed on a side of the pair of rails **120** facing each other. The transport belt **208** forms a supporting surface **210** extending in the X-axis direction, and supporting surfaces **210** on the pair of rails **120** collectively support the substrate **110**. Each rail **120** also includes a blocking portion **220** located above the supporting surface **210**, and the blocking portion **220** is formed by a baffle disposed at the top of the rail **120**. When the substrate **110** is located on the supporting surface **210** of the pair of rails **120**, the substrate **110** is located in its transport position where the substrate **110** can be transported in the X-axis direction. When the substrate **110** is raised along the Z-axis direction to contact the blocking portion **220**, the substrate **110** is located in its dispensing position where the substrate **110** is proximate the dispensing unit **105**, where the dispensing unit **105** is capable of dispensing a viscous material in a desired position on the substrate **110**.

[0027] Still as shown in FIG. 2, the heating device **201** is located below the rail **120** and the drive device **203** is located below the heating device **201**. The heating device **201** is used to heat the substrate **110**, and the drive device **203** drives the heating device **201** to rise and fall along the Z-axis direction to enable the heating device **201** to be proximate or remote from the substrate **110**. The heating device **201** includes a wind deflector **202** located at the top thereof, and the wind deflector **202** has uniformly distributed air guide holes for directing hot air to blow evenly towards the substrate **110**. The drive device **203** is, for example, an air cylinder. Each rail **120** is mounted

with one lifting device **250** located below the substrate **110** so as to be able to push the substrate **110** to rise. The lifting device **250** is capable of lifting the substrate **110** from the transport position to the dispensing position under pushing of the heating device **201** and is capable of causing the substrate to fall from the dispensing position to the transport position when the heating device **201** is descends.

[0028] FIGS. **3A-3E** show a specific structure of the lifting device **250**, where FIG. **3A** is a stereoscopic view of the rail **120** and the lifting device **250**, FIG. **3B** is an exploded view of the rail **120** and the lifting device **250**, and FIG. **3C** is a partial enlarged view of a top view of the rail **120** and the lifting device **250**, FIG. **3D** is a cross-sectional view along line A-A in FIG. **3C**, and FIG. **3E** is a cross-sectional view along line B-B in FIG. **3C**. As shown in FIGS. **3A-3E**, the lifting device **250** includes a lifting member **310**, a reset member **320**, and a connecting device **330**, and the lifting member **310** is movably connected to the rail **120** by the connecting device **330** and is movable in the Z-axis direction. The reset member **320** is disposed between the lifting member **310** and the rail **120** to be compressed as the lifting member **310** rises relative to the rail **120** and to be able to lower the lifting member **310** relative to the rail **120** by a resilient force. The lifting member **310** is located below the substrate **110**, and is capable of moving the substrate **110** from the transport position to the dispensing position when it is raised, and moving the substrate **110** from the dispensing position to the transport position when it descends. The connecting device **330** includes a connecting slot **332** disposed on the lifting member **310** and a connecting member **334** connected to the rail **120**, and the connecting member **334** is, for example, a bolt.

[0029] The lifting member **310** is clevis-like and extends by a distance along an extension direction of the rail **120** (i.e., in the X-axis direction). The lifting member **310** includes a pair of side plates **312** and a base plate **314** connected between the pair of side plates **312**. Together, the pair of side plates **312** and the base plate **314** form the clevis-like lifting member **310** that enables the lifting member **310** to be sleeved on the rail **120** from below the rail **120**, such that the pair of side plates **312** are located on opposite sides of the rail **120**, while the base plate **314** is located below the rail **120**. The pair of side plates **312** of the lifting member **310** each have several connecting slots **332** on each side plate **312**, the several connecting slots **332** are arranged along the extension direction (i.e., the X-axis direction) of the lifting member **310**, and each connecting slot **332** extends along the Z-axis direction, and thus has a length to allow movement of the lifting member **310** along the Z-axis direction.

[0030] The lifting device **250** also includes several guide wheels **335**, and the quantity of guide wheels **335** is corresponding to the quantity of connecting slots **332**. The guide wheel **335** is sleeved on the connecting member **334** which is secured to the rail **120**, and both the guide wheel **335** and the connecting member **334** are inserted into the connecting slot **332**. The guide wheel **335** is configured to cooperate with the connecting slot **332** to guide the lifting member **310** to move smoothly in the Z-axis direction. The outer surface of the guide wheel **335** is set as a slit **338** that extends circumferentially along the guide wheel **335**, and the width of the slit **338** is slightly greater than the thickness of the side plate **314** of the lifting member **310**, so that a pair of opposite edges **337** of the side plate **314** forming the connecting slot **332** are clamped in the slit **338** of the guide wheel **335**.

[0031] As shown in FIGS. **3B** and **3D**, the side plate **314** is provided with an opening **339** that is enlarged in size compared to the width of the connecting slot **332** at the bottom end of the connecting slot **332**, the opening **339** is in communication with the connecting slot **332**, and the opening **339** is sized to be larger than the outer diameter of the guide wheel **335**, so that the guide wheel **335** can be inserted into the connecting slot **332** via the opening **339**, and after moving upward, be clamped by its slit **338** to form a pair of edges **337** of the slit **338**. As such, the guide wheel **335** is capable of directing the lifting member **310** to move smoothly in the Z-axis direction.

[0032] In some examples, the reset member **320** is a helical spring. There may be a plurality of reset members **320**, which are arranged along the extension direction (i.e., the X-axis direction) of

the lifting member **310**. The reset member **320** is disposed in a hole of the rail **120** with the bottom end of the rail **120** extending through the hole of the rail **120** to abut against the bottom plate **314** of the lifting member **310**, and with the top end thereof connected to the rail.

[0033] The lifting device **250** also includes a guide pin **325**, and the quantity of guide pins **325** is the same as the quantity of reset members **320**. A bottom end **326** of the guide pin **325** is connected to the bottom plate **314** of the lifting member **310**, and the guide pin **325** is located between the pair of side plates **312**. The guide pin **325** extends generally along the Z-axis direction, and the reset member **320** is sleeved on the guide pin **325** such that the reset member **320** can be compressed or released along the Z-axis direction.

[0034] As the lifting member **310** rises along the Z-axis relative to the rail **120**, the reset member **320** is compressed by the lifting member **310** to accumulate a resilient force. The lifting member **310** can also descend along the Z-axis relative to the rail **120** under the action of the resilient force of the reset member **320**.

[0035] FIGS. **4A-4C** show a status change of the lifting device **250** of the dispenser **100** as the substrate **110** moves between the transport position and the dispensing position. FIG. **4A** is a cross-sectional view of the substrate **110** in the transport position in FIG. **2**, FIG. **4B** is a cross-sectional view of the heating device **201** when raised to contact the lifting device **250**, and FIG. **4C** is a cross-sectional view of the substrate **110** when raised to the dispensing position.

[0036] As shown in FIG. **4A**, when the substrate **110** is in the transport position, the substrate **110** is supported by a supporting surface **210** of the pair of rails **120**, and the lifting member **310** of the lifting device **250** does not contact, but rather is spaced apart from, the wind deflector **202** of the heating device **201**. When there is a need to dispense a viscous material onto the substrate **110**, the heating device **201** is driven upward by the drive device **203** so as to bring the heating device **210** into proximity to the substrate **110** to perform a heating operation on the substrate **110**, while also being able to move the substrate **110** to the dispensing position through the heating device **210**.

[0037] As shown in FIG. **4B**, the heating device **201** is driven by the drive device **203** to raise such that its wind deflector **202** is in contact with the lifting member **310** of the lifting device **250**. Thereafter, the drive device **203** continues to drive the heating device **201** to move axially along the Z-axis, and movement of the heating device **201** will cause the lifting member **310** of the lifting device **250** to rise relative to the rail **120**. As the lifting member **310** is located below the substrate **110**, the lifting member **310** will contact and push the substrate **110** to rise until it reaches the position shown in FIG. **4C**.

[0038] As shown in FIG. **4C**, when the substrate **110** rises to abut against the blocking portion **220**, the substrate **110** is blocked from continuing to rise and the substrate **110** reaches its dispensing position. During movement from the state shown in FIG. **4B** to the state shown in FIG. **4C**, the movement of the lifting member **310** relative to the rail **120** compresses the reset member **320**.

[0039] When the dispensing operation for the substrate **110** is complete, the drive device **203** drives the heating device **201** to descend along the Z-axis, and the lifting member **310** is able to descend along the Z-axis under the resilient force of the reset member **320** as it is no longer pushed by the heating device **201**, causing the substrate **110** to also descend until the substrate **110** is supported by the supporting surface **210** of the rail **120** to reach its transport position.

[0040] The dispenser **100** usually needs to be designed to process substrates **110** with different widths. In order to move the substrate **110** between the transport position and the dispensing position, the distance between the lifting devices **250** needs to adapt to the width of the substrate **110** so that the lifting device **250** can push the substrate **110** to lift or descend, and the distance between the pair of rails **120** need to adapt to the width of the substrate **110** so as to support the substrate **110**. As the contributor **100** of the present disclosure is provided with lifting devices **250** mounted on the pair of rails **120** respectively, the lifting devices **250** can be pushed by the heating device **201** and further push the substrate **110** to move, so that the substrate **110** can move from its transport position to the contribution position to perform the contribution operation. When the

distance between the pair of rails **120** needs to be adjusted according to the width of the substrate **110**, the distance between the lifting devices **250** mounted on the pair of rails **120** is also adjusted accordingly, so that the distance between the lifting devices **250** also adapts to the width of the substrate **110**. As such, the same lifting device **110** and heating device **201** can adapt to different substrate widths.

[0041] In addition, as the lifting device **250** of the present disclosure surrounds the rail **120** by using the lifting member **310**, when the heating device **201** contacts the lifting member **310** to push the lifting member upward and perform a heating operation on the substrate **110**, the lifting member **310** can isolate the rail **120** from the heating device **210** to protect the rail **120** (especially the transport belt **208** on the rail **120**) from high temperature baking from the heating device **210**.

[0042] Although the present disclosure has been described in connection with examples outlined above, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or foreseeable now or in the near future, may be apparent to those having at least ordinary skill in the art. In addition, the technical effects and/or technical problems described in the present specification are exemplary and not limiting; therefore, the disclosure in the present specification may be used to solve other technical problems and have other technical effects and/or may solve other technical problems. Therefore, examples of the present disclosure as set forth above are intended to be illustrative and not limiting. Various changes may be made without departing from the spirit or scope of the present disclosure. Therefore, the present disclosure is intended to include all known or earlier developed alternatives, modifications, variations, improvements and/or substantial equivalents.

Claims

1. A dispenser (**100**) configured to dispense a material on a substrate (**110**), wherein the dispenser (**100**) comprises: a pair of rails (**120**) configured to input and output the substrate (**110**) to the dispenser (**100**) in an X-axis direction, wherein one of the pair of rails (**120**) is movable in a Y-axis direction towards or away from the other to adjust a distance between the pair of rails (**120**), a supporting surface (**210**) and a blocking portion (**220**) higher than the supporting surface (**210**) are disposed on each of the pairs of rails (**120**), the supporting surface (**210**) of the pair of rails (**120**) is configured to support the substrate (**110**) in a transport position, and the blocking portion (**220**) of the pair of rails (**120**) is configured to keep the substrate (**110**) in a dispensing position; a heating device (**201**) located below the pair of rails (**120**) in a Z-axis direction and movable in the Z-axis direction; and a pair of lifting devices (**250**) mounted to the pair of rails (**120**) respectively, each of the pair of lifting devices (**250**) comprising: a lifting member (**310**) movably connected to the rail (**120**) and movable in the Z-axis direction; and a reset member (**320**) located between the rail (**120**) and the lifting member (**310**) to be compressed when the lifting member (**310**) is raised relative to the rail (**120**) and to be able to lower the lifting member (**310**) relative to the rail (**120**) by a resilient force; wherein the lifting member (**310**) is configured to be able to be raised under pushing of the heating device (**201**) to cause the substrate (**110**) to rise from the transport position to the dispensing position, and to lower the substrate (**110**) to the transport position from the dispensing position under an action of the reset member (**320**) when the heating device (**201**) descends.
2. The dispenser (**100**) according to claim 1, wherein: the lifting device (**250**) further comprises a connecting device (**330**) comprising a connecting slot (**332**) disposed on the lifting member (**310**) and a connecting member (**334**) connected to the rail (**120**), the connecting slot (**332**) extending along the Z-axis direction, the connecting member (**334**) being inserted into the connecting slot (**332**) and being movable in the connecting slot (**332**) in the Z-axis direction.
3. The dispenser (**100**) according to claim 2, wherein: the lifting member (**310**) is a clevis-like and sleeved on the rail (**120**) from the underside of the rail (**120**), the lifting member (**310**) comprising a pair of side plates (**312**) and a base plate (**314**) connected between the pair of side plates (**312**), the

- pair of side plates (312) being respectively located on opposite sides of the rail (120), the base plate (314) being located below the rail (120), wherein the pair of side plates (312) each has at least one connecting slot (332).
4. The dispenser (100) according to claim 3, wherein: the lifting member (310) extends by a distance along the X-axis direction; and the lifting device (250) comprises several reset members (320), several connecting slots (332) disposed on each of the side plates (312), and several connecting members (334) corresponding to the several connecting slots (332), wherein the several reset members (320) are arranged in the X-axis direction, and the several connecting slots (332) are arranged in the X-axis direction.
5. The dispenser (100) according to claim 4, wherein: a hole (340) is provided in the rail (120), and the reset member (320) is at least partially located in the hole (340).
6. The dispenser (100) according to claim 1, wherein: the reset member (320) is a helical spring.
7. The dispenser (100) according to claim 6, wherein: the dispenser (100) also comprises a guide pin (325), a bottom end (326) of the guide pin (325) is connected to the lifting member (310), and the reset member (320) is sleeved on the guide pin (325).
8. The dispenser (100) according to claim 3, wherein: the lifting device (250) further comprises a guide wheel (335) having a slit (338) on an exterior surface thereof, the slit (338) extending circumferentially along the guide wheel (335); wherein the guide wheel (335) is sleeved on the connecting member (334), the guide wheel (335) is inserted into the connecting slot (332), and opposite edges (337) of the side plate (312) of the lifting member (310) forming the connecting slot (332) are clamped in the slit (338) of the guide wheel (335).
9. The dispenser (100) according to claim 1, wherein: the blocking portion (220) is formed by a baffle mounted at a top end of the rail (120).
10. The dispenser (100) according to claim 1, further comprising: a drive device (203) disposed below the heating device (201) for driving movement of the heating device (201) in the Z-axis direction.
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